Divisibility and congruences

Lecture and worked examples

Objectives

Use divisibility in $\mathbb{Z}$ and its properties; carry out a Euclidean division; compute with congruences; establish a divisibility test; study remainders of powers.

Introductory activity

17 October 2026 is a Saturday. The days of the week are numbered from 0 (Saturday) to 6 (Friday): Sunday has number 1, Monday number 2, and so on.

Introductory activity (2)

1. What day of the week will it be 7 days later? 14 days? 700 days?
2. What day will it be 100 days later? Explain how to get the answer without counting the days one by one.
3. Same question for 1,000 days.
4. The year 2027 is not a leap year. What day of the week will 17 October 2027 be?
5. 1 January 2027 comes 76 days after 17 October 2026. What day will it be?
6. Two durations a and b, in days, lead to the same day of the week. What can be said about $a - b$? State a general condition.

Introductory activity (3)

Observing is not proving: the properties conjectured here are established in the course.

Divisibility in $\mathbb{Z}$

Let a and b be two integers. We say that a *divides* b, written $a \mid b$, if there is an integer k such that $b = ka$. We also say that a is a *divisor* of b, or that b is a *multiple* of a.

Divisibility in $\mathbb{Z}$

Let a, b, c be integers.

1. If $a \mid b$ and $b \mid c$, then $a \mid c$.
2. If $a \mid b$ and $a \mid c$, then $a \mid bu + cv$ for all integers u and v.
3. If $a \mid b$ and $b \neq 0$, then $|a| \leq |b|$.

Divisibility in $\mathbb{Z}$

1. $b = ka$ and $c = k'b$ give $c = (k'k)a$. 2. $b = ka$ and $c = k'a$ give $bu + cv = (ku + k'v)a$. 3. $b = ka$ with $k \neq 0$, so $|k| \geq 1$ and $|b| = |k|\,|a| \geq |a|$.

Divisibility in $\mathbb{Z}$

Let $a \in \mathbb{Z}$ and $b \in \mathbb{N}$, $b \geq 1$. There is a unique pair of integers (q, r) such that $a = bq + r$ and $0 \leq r < b$. We call q the *quotient* and r the *remainder*.

Existence. Let $q = \left\lfloor \frac{a}{b} \right\rfloor$, the largest integer less than or equal to $\frac{a}{b}$. Then $q \leq \frac{a}{b} < q+1$, so $bq \leq a < bq + b$, and $r = a - bq$ satisfies $0 \leq r < b$.

Uniqueness. If $a = bq + r = bq' + r'$ with $0 \leq r, r' < b$, then $b(q - q') = r' - r$ and $-b < r' - r < b$. The only multiple of b strictly between $-b$ and b is 0, so $r = r'$ and then $q = q'$.

The remainder is always nonnegative: $-47 = 5 \times (-10) + 3$, so the division of -47 by 5 has quotient -10 and remainder 3, not -9 and -2.

Congruences

Let $n \geq 2$ be an integer. Two integers a and b are congruent modulo n, written $a \equiv b \pmod{n}$, if $n \mid a - b$.

Congruences

$a \equiv b \pmod{n}$ if and only if a and b have the same remainder in the Euclidean division by n. In particular, every integer is congruent modulo n to its remainder, which belongs to $\{0, \ldots, n-1\}$.

Congruences

Write $a = nq + r$ and $b = nq' + r'$ with $0 \leq r, r' < n$. Then $a - b = n(q - q') + (r - r')$ and $-n < r - r' < n$. So $n \mid a - b$ if and only if $n \mid r - r'$, if and only if $r = r'$.

Congruences

If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then

$$a + c \equiv b + d, \qquad a - c \equiv b - d, \qquad ac \equiv bd, \qquad a^k \equiv b^k \pmod{n}$$

for every natural number k. The relation is also transitive: if $a \equiv b$ and $b \equiv c \pmod{n}$, then $a \equiv c \pmod{n}$.

Congruences

Sum: $(a+c)-(b+d)=(a-b)+(c-d)$ is a sum of two multiples of n. Product: $ac-bd=a(c-d)+d(a-b)$, a multiple of n. Power: induction on k from the product. Transitivity: $a-c=(a-b)+(b-c)$.

Congruences

A congruence is not simplified like an equality: $2 \times 3 \equiv 2 \times 1 \pmod 4$, but $3 \not\equiv 1 \pmod 4$.
Division is possible only in specific cases (chapter 7).

Congruences

Look for a power congruent to 1 (or to -1), then use the Euclidean division of the exponent. Example: $2^3 = 8 \equiv 1 \pmod 7$ and $2{,}026 = 3 \times 675 + 1$, so $2^{2026} = \left(2^3\right)^{675} \times 2 \equiv 1^{675} \times 2 = 2 \pmod 7$.

Congruences

To study an expression modulo n, draw up a table giving its value modulo n for each possible remainder of x. Example: modulo 4, $x \equiv 0, 1, 2, 3$ gives $x^2 \equiv 0, 1, 0, 1$. A square is therefore congruent to 0 or 1 modulo 4.

Congruences

If $N = \sum_{k=0}^{p} a_k 10^k$ is written with the digits $a_0, \ldots, a_p$, since $10 \equiv 1 \pmod 9$, we have $10^k \equiv 1$ and $N \equiv a_0 + a_1 + \cdots + a_p \pmod 9$. An integer is divisible by 9 if and only if the sum of its digits is.